ECE 6256 Assignment 2: SFT-LoRA and Direct Preference Optimization

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Objective and Data Generation

The objective was to improve a small instruction-tuned model's ability to request clarification when a user's information is insufficient, while continuing to answer sufficiently specified requests directly and preserving general knowledge. The base model was Llama-3.2-1B-Instruct. The preference dataset contained 2,110 prompt/completion pairs: 1,035 ambiguous (49.1%) and 1,075 clear (50.9%). Each ambiguous example paired a specific clarifying question with an unsupported assumption; each clear example paired a direct response with an unnecessary follow-up question. Examples spanned arithmetic, Python output, text transformation, logical reasoning, general knowledge, writing, planning, and document extraction. The dataset was generated by GPT-6.1 luna and used prompts in *generate_preference_data.py*. A 100-pair hand-verified file recorded 91 accepts and 9 rejects (9% rejection rate). One reject contained a chosen clarification unrelated to its prompt; eight concerned weak or mismatched negative questions. They were reviewed for correct ambiguity classification, appropriate chosen responses, and meaningful rejected alternatives.

Training Method

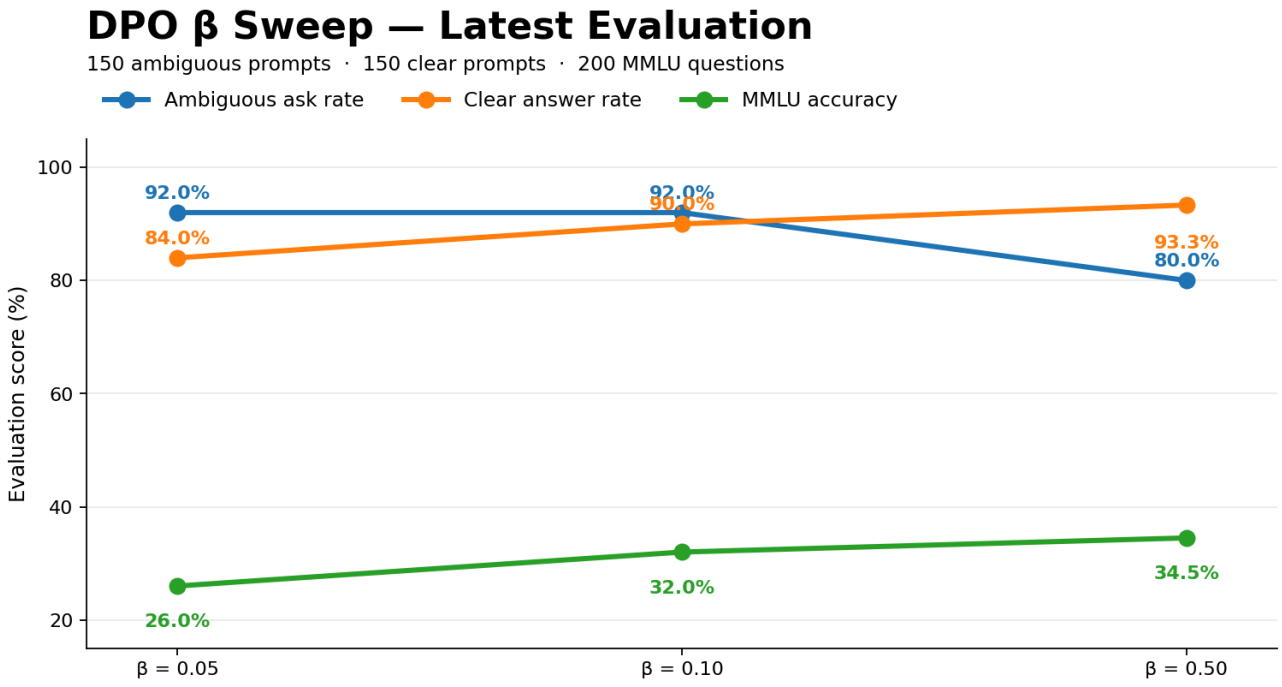
- **Base:** `meta-llama/Llama-3.2-1B-Instruct`.
- **SFT-LoRA:** 2 epochs, 528 optimizer steps per epoch on 2,110 preferred completions; mean loss **2.2908** → **1.8154**.
- **DPO:** separate LoRA runs at `β = 0.05`, `0.1`, and `0.5`; 1 epoch each, **1,055 steps** per run, batch size 2, learning rate `1e-6`, maximum sequence length 1,024. Reference scores cached with original base adapter disabled, according to the logs.
- **Evaluation:** local development sets of 150 ambiguous, 150 clear, and 200 MMLU prompts; ask / answer / other classifications by `meta-llama/Llama-3.1-8B-Instruct`, judge rubric v1.0, seed 6604

```
loss = model(input_ids=input_ids, labels=labels).loss # SFT objective
loss = -F.logsigmoid(beta * margin).mean() # DPO Objective
```

Latest Evaluation Results

Checkpoint	Ambiguous ask	Clear answer	MMLU
Base	61.33%	94.67%	32.0%
SFT	79.33%	91.33%	32.0%
DPO β=0.05	92.00%	84.00%	26.0%
DPO β=0.10	**92.00%**	**90.00%**	**32.0%**
DPO β=0.50	80.00%	93.33%	34.5%

Best local checkpoint: DPO β=0.10



Base model: 61.3% ask · 94.7% answer · 32.0% MMLU

β = 0.10 meets all three behavioral and retention targets on this development set.

The β sweep revealed a trade-off between clarification and direct answering. At **β = 0.05**, the model achieved a 92% ask rate, but its answer rate dropped to 84% and MMLU accuracy fell to 26%. At **β = 0.1**, the ask rate remained 92%, while answer rate improved to 90% and MMLU recovered to 32%. At **β = 0.5**, answer rate rose to 93.33%, but ask rate declined to 80%. **Over-asking** appeared at β = 0.05 when the model requested an already-provided sentence instead of rewriting it. **Capability degradation** also appeared at β = 0.05 when the model produced an incomplete answer to an MMLU biology question. This suggests possible shortcut learning rather than proven catastrophic forgetting or reward hacking. Overall, **β = 0.1 provided the best balance**, meeting all three evaluation targets.

The failures were observed after training, so their exact onset during optimization is unknown.

Three Qualitative Transcripts

1. Successful Clarification (β=0.1)
Prompt: "Give me a one-paragraph summary of chapter 6 of my manuscript."
Base: "I'd be happy to help, but I don't have any information..." *(excerpt)*
SFT: "Could you share the chapter 6 excerpt from your manuscript?"
DPO 0.1: "Can you provide the excerpt from chapter 6 of your manuscript?"
Analysis: The base model recognized the missing context but produced an unnecessarily lengthy response with unrelated suggestions. SFT and DPO both generated concise, targeted clarification questions. This demonstrates improved response efficiency without sacrificing accuracy or making unsupported assumptions.

2. Over-Asking Failure (β=0.05)
Prompt: "Rewrite this sentence more politely: 'Send me the file now.'"
Base: "Could you please send me the file as soon as possible?" *(excerpt)*
DPO 0.05: "Could you please provide the sentence you'd like rewritten?"
DPO 0.1: "Could you please send the file to me as soon as possible?"
Analysis: Despite receiving a complete request, β=0.05 unnecessarily asked for the original sentence again. This suggests the model developed an excessive preference for clarification rather than evaluating whether information was actually missing. β=0.1 corrected this behavior and responded directly.

3. Capability-Retention Failure (β=0.05)
Prompt (abridged): Which process involves gas exchange in earthworms, grasshoppers, and alveoli? (Correct: B — Respiration)
Base/SFT/DPO 0.1: Correctly answered B.
DPO 0.05: "To determine the correct answer, let's"
Analysis: Unlike the other models, β=0.05 failed to produce the required answer despite the question being fully specified. This illustrates degraded task completion and is consistent with its lower MMLU accuracy (26% versus 32% for the base model). However, the incomplete response does not conclusively prove catastrophic forgetting, since the failure could also involve response generation or formatting.

AI Use Disclosure

I used ChatGPT to help generate and diversify prompts and preference pairs for the training dataset. I wrote the initial implementations of `train_sft.py` and `train_dpo.py`, then used AI assistance to optimize the code, debug errors, and troubleshoot training issues. AI was also used to generate the β-sweep graph, develop scripts to extract actual model-generated transcripts, and help me understand SFT, DPO, and the evaluation results. Additionally, I used AI to review dataset quality and assist with drafting and refining the report's analysis. The reported performance metrics came from my local training and evaluation runs on PACE-ICE.

Appendix

Out of 100 sampled preference pairs, **9 were flagged for rejection (9%)** during review. Most of the issues came from unrealistic clarification questions rather than incorrect answers.

Pair ID	Reason for Rejection
`001296`	Asked about rounding when the finish time was straightforward.
`001199`	Asked for an unrelated example instead of finding a meeting slot.
`000075`	Asked about an email closing for a blog outline.
`001375`	The chosen response asked about an unrelated opening requirement instead of the missing feedback.
`000577`	Unnecessary question about rounding walking times.
`000334`	Unnecessary rounding question for simple subtraction.
`000222`	Asked for a conceptual example instead of checking seating capacity.
`001575`	Compared a poem's length to an email.
`001885`	Asked about a scientific analogy for an event description.